

DBMS: ASSIGNMENT 7

MongoDB- CRUD Operations

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Case Study: Staff Performance Document Store

```
test> use staffPerformanceHub
switched to db staffPerformanceHub
```

```
staffPerformanceHub> db.staff.insertMany([
| { Id: 1, Name: "Adi Maitre", Salary: 10000, Rating: 5, status: "active" },
| { Id: 2, Name: "Kritika Chaudhari", Salary: 15000, Rating: 4, status: "active" },
| { Id: 3, Name: "Ritobrata Baksi", Salary: 20000, Rating: 3, status: "active" },
| { Id: 4, Name: "Sarang Kolekar", Salary: 18000, Rating: 2, status: "inactive" }
| ])
{
  acknowledged: true
}
```

```
staffPerformanceHub> db.staff.find({ status: "active" })
```

```
staffPerformanceHub> db.staff.findOne({ Name: "Adi Maitre" })
```

```
staffPerformanceHub> db.staff.find({ Rating: 5, status: "active" })
```

```
staffPerformanceHub> db.staff.find({
| $or: [
|   { Rating: 5 },
|   { Salary: { $gt: 18000 } }
| ]
| })
```

```
staffPerformanceHub> db.staff.find({ status: { $ne: "inactive" } })
```

```
staffPerformanceHub> db.staff.find({ Salary: { $gt: 15000 } })
```

```
staffPerformanceHub> db.staff.updateOne(
| { Rating: 5 },
| { $set: { Salary: 20000 } }
| )
```

```

staffPerformanceHub> db.staff.updateMany(
| { Rating: { $gte: 4 } },
| { $inc: { Salary: 5000 } }
| )

staffPerformanceHub> db.staff.deleteOne({ Id: 4 })

staffPerformanceHub> db.staff.find().sort({ Salary: -1 })

staffPerformanceHub> db.staff.find().limit(2)

staffPerformanceHub> db.staff.aggregate([
| {
|   $group: {
|     _id: null,
|     totalSalary: { $sum: "$Salary" }
|   }
| }
| ])
[ { _id: null, totalSalary: 65000 } ]

```

Project: Smart Tax Portal Document DB

test> use smartTaxInsight

switched to db smartTaxInsight

```

smartTaxInsight> db.user_login.insertMany([
| { user_id: 1, first_name: "Adi", last_name: "Maitre", role: "taxpayer", status: "active" },
| { user_id: 2, first_name: "Kritika", last_name: "Chaudhari", role: "taxpayer", status: "active" },
| { user_id: 3, first_name: "Ritobrata", last_name: "Baksi", role: "taxpayer", status: "active" },
| { user_id: 4, first_name: "Sarang", last_name: "Kolekar", role: "taxpayer", status: "inactive" }
| ])

```

```

smartTaxInsight> db.taxpayer_profile.insertMany([
| { taxpayer_id: 1, user_id: 1, pan: "PAN001", city: "Pune" },
| { taxpayer_id: 2, user_id: 2, pan: "PAN002", city: "Mumbai" },
| { taxpayer_id: 3, user_id: 3, pan: "PAN003", city: "Delhi" },
| { taxpayer_id: 4, user_id: 4, pan: "PAN004", city: "Nashik" }
| ])

```

```

smartTaxInsight> db.income_sources.insertMany([
| { taxpayer_id: 1, amount: 600000 },
| { taxpayer_id: 2, amount: 300000 },

```

```
| { taxpayer_id: 3, amount: 800000 },  
| { taxpayer_id: 4, amount: 200000 }  
| ])
```

```
smartTaxInsight> db.deductions.insertMany([  
| { taxpayer_id: 1, amount: 50000 },  
| { taxpayer_id: 2, amount: 20000 },  
| { taxpayer_id: 3, amount: 60000 },  
| { taxpayer_id: 4, amount: 10000 }  
| ])
```

```
smartTaxInsight> db.user_login.find({ status: "active" })
```

```
smartTaxInsight> db.user_login.findOne({ first_name: "Adi" })
```

```
smartTaxInsight> db.user_login.updateOne(  
| { user_id: 4 },  
| { $set: { status: "active" } }  
| )
```

```
smartTaxInsight> db.income_sources.updateMany(  
| { amount: { $gt: 500000 } },  
| { $inc: { amount: 10000 } }  
| )
```

```
smartTaxInsight> db.deductions.deleteOne({ taxpayer_id: 4 })
```

```
smartTaxInsight> db.income_sources.find().sort({ amount: -1 })
```

```
smartTaxInsight> db.income_sources.find().skip(1).limit(2)
```

```
smartTaxInsight> db.user_login.aggregate([  
| {  
|   $lookup: {  
|     from: "taxpayer_profile",  
|     localField: "user_id",  
|     foreignField: "user_id",  
|     as: "profile"  
|   }  
| }  
| ])
```

```
smartTaxInsight> db.income_sources.aggregate([  
| {
```

```
| $group: {  
|   _id: "$taxpayer_id",  
|   total_income: { $sum: "$amount" }  
| }  
| },  
| {  
|   $match: { total_income: { $gt: 500000 } }  
| },  
| {  
|   $project: {  
|     total_income: 1,  
|     tax: { $multiply: ["$total_income", 0.10] }  
|   }  
| }  
| )
```